12TH CHEMISTRY · CHAPTER 20 · FULL CHAPTER NOTES

Aromatic Hydrocarbons

Topic-wise short and long questions with answers from the complete chapter

INTRODUCTION

Q.1 [Short] What are aromatic hydrocarbons (arenes)? Why were they called "aromatic"?

Benzene, C_6H_6, is the simplest aromatic hydrocarbon. Benzene derived products are well known to be pleasantly fragrant, so organic compounds containing benzene rings were classified as "aromatic" (sweet smelling) by scientists in the early 19th century, e.g., vanilla. Organic hydrocarbons containing one or more benzene rings are known as **arenes**. The term "aromatic" is still used today for molecules with benzene and benzene-like rings in their structures, even if they are not good-smelling but have foul smells.

Q.2 [Short] Is it correct to say that only aromatic compounds smell good and sweet?

No. Non-aromatic compounds can also have a strong smell. For example, camphor, extracted from the camphor laurel tree, releases a strong, minty aroma, yet it has no benzene ring in its structure. Also, many compounds with benzene rings have foul smells.

20.1 STRUCTURE OF BENZENE

Q.3 [Short] Describe the Kekule structure of benzene. What was its weakness?

After its discovery the molecular formula of benzene was found to be C_6H_6. Some scientists presented linear structures, but these were rejected as they could not explain the properties of benzene. In 1865 the German professor August Kekule elucidated it as a six-membered cyclic compound having alternate single and double bonds. This structure explained many properties of benzene successfully, but it did not fully explain all its physical and chemical properties.



Kekulé structures delocalised structure

Q.4 [Short] What do X-ray studies tell us about the structure of benzene?

X-ray studies show that benzene has a planar configuration: a regular hexagon of six sp^2 hybridized carbon atoms. Each carbon is connected to three other atoms with bond angles of 120° for C–C–C and H–C–C. The C–H bond length is 109 pm. The C–C bond length is 139 pm, which falls between the C–C single bond length (154 pm) and the C=C double bond length (133 pm). All six C–C bonds are equal.

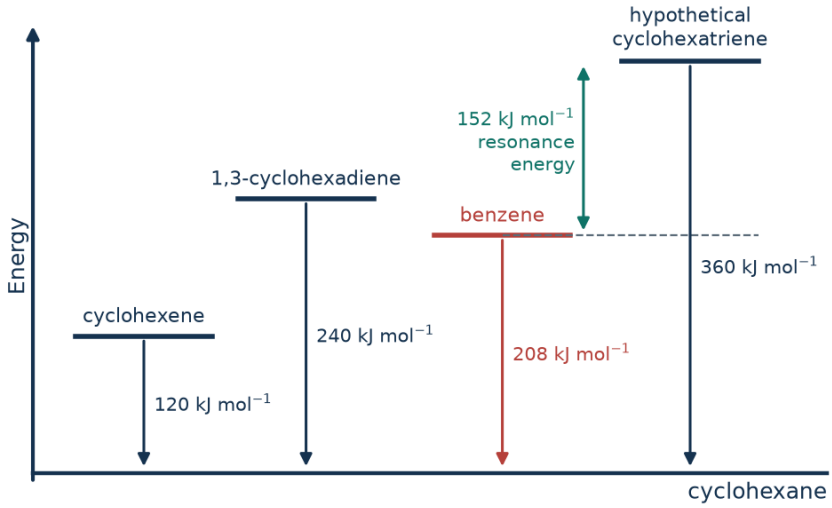
Bond	Bond length
C–C	154 pm
C=C	133 pm
C≡C	120 pm
C–C in benzene	139 pm

20.1.1 Thermodynamic Stability of Benzene and Resonance

Q.5 [Long] Explain the thermodynamic stability of benzene using heats of hydrogenation. Define resonance energy.

Benzene, like alkenes, has double bonds, but it is much more stable than alkenes. The evidence comes from heats of hydrogenation, with cyclohexane as the common product and reference point:

- Cyclohexene → cyclohexane releases 120 kJ mol^{-1}.
- So 1,3-cyclohexadiene is expected to release 240 kJ mol^{-1} and the hypothetical 1,3,5-cyclohexatriene 360 kJ mol^{-1}.
- Benzene actually releases only 208 kJ mol^{-1}, i.e., 152 kJ mol^{-1} less than the theoretical value.



Energy level diagram of hydrogenation of cyclohexenes and benzene

So benzene is much more stable than the hypothetical cyclohexatriene. This stability is explained by the delocalisation of the π electrons over all six carbon atoms. The energy difference of 152 kJ mol^{-1} is called the **resonance energy (delocalisation energy)** of benzene.

The textbook also prints "158.5 kJ mol^{-1} lower than expected" in one sentence; 360 – 208 = 152 kJ mol^{-1}, the value it uses for the resonance energy.

Q.6 [Short] Define resonance energy of benzene.

It is the difference between the theoretical heat of hydrogenation of the hypothetical 1,3,5-cyclohexatriene (360 kJ mol^{-1}) and the actual heat of hydrogenation of benzene (208 kJ mol^{-1}), i.e., 152 kJ mol^{-1}. It measures the extra stability of benzene due to the delocalisation of its π electrons.

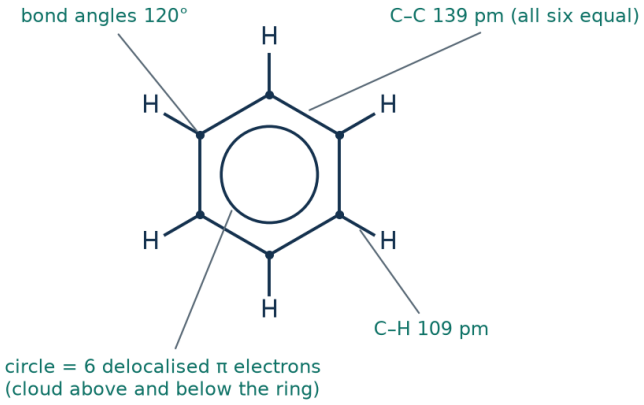
20.1.2 Atomic Orbital Structure of Benzene

Q.7 [Long] Explain the shape of the benzene molecule on the basis of the atomic orbital (hybridization) approach.

σ framework: each carbon atom in benzene is sp^2 hybridized. The three sp^2 hybrid orbitals of each carbon form three σ bonds: two with the adjacent carbon atoms (sp^2–sp^2) and one with a hydrogen atom (sp^2–s). All the sp^2 orbitals lie in the same plane, so all carbon and hydrogen atoms are coplanar with angles of 120°.

π system: the six unhybridized 2p orbitals, one on each carbon, are perpendicular to this σ framework and parallel to each other. They overlap sideways on both sides to form a π-electron cloud above and below the plane of the ring, making it a conjugated molecule. The six electrons are shared equally among all six carbons; this is called **delocalization** of π electrons.

Delocalization makes all six C–C bond lengths the same (139 pm). So benzene is represented by a hexagon with an inscribed circle, where the circle represents the delocalised π electrons.



Bonding in benzene

Q.8 [Short] Why is benzene represented by a hexagon with a circle inside it?

The delocalisation of the six π electrons makes all six C–C bonds the same length (139 pm), so there are no fixed single and double bonds. The circle represents the delocalised π electrons.

Quick Check 20.1

Q.9 [Short] **a) The delocalization (resonance) energy of benzene is about 152 kJ mol^{-1}. Explain what that means and how it affects the reactivity of benzene.**

It means that benzene is 152 kJ mol^{-1} more stable (lower in energy) than the hypothetical 1,3,5-cyclohexatriene with three separate double bonds: its heat of hydrogenation is 208 kJ mol^{-1} instead of the expected 360 kJ mol^{-1}. This extra stability comes from the delocalisation of the π electrons. As a result benzene is less reactive than alkenes: it resists addition reactions, which would destroy the delocalisation, and undergoes electrophilic substitution instead.

Q.10 [Short] **b) The X-ray studies show that benzene has a planar structure. If benzene were not planar, what difference in the reactivity of benzene would be seen?**

If the ring were not planar, the six 2p orbitals would not be parallel and could not overlap sideways all round the ring. The π electrons would not be delocalised, and there would be no resonance stabilisation. Benzene would then have ordinary localised double bonds and would react like an alkene: it would be much more reactive and would undergo addition reactions easily.

Q.11 [Short] **c) Why all the bonds in the benzene ring are of equal bond length?**

The six π electrons are delocalised and shared equally among all six carbon atoms, so there are no separate single and double bonds. Every C–C bond is identical, with a length of 139 pm, between a single bond (154 pm) and a double bond (133 pm).

Q.12 [Short] **d) How the atomic orbital concept explains the delocalization of π electrons in benzene?**

Each carbon is sp^2 hybridized and uses its three hybrid orbitals for σ bonds in one plane. Each carbon is left with one unhybridized 2p orbital, perpendicular to the ring and parallel to the others. Each p orbital overlaps sideways with the p orbitals on both sides, forming a continuous π-electron cloud above and below the plane of the ring. The six electrons are therefore shared equally by all six carbons, i.e., delocalised.

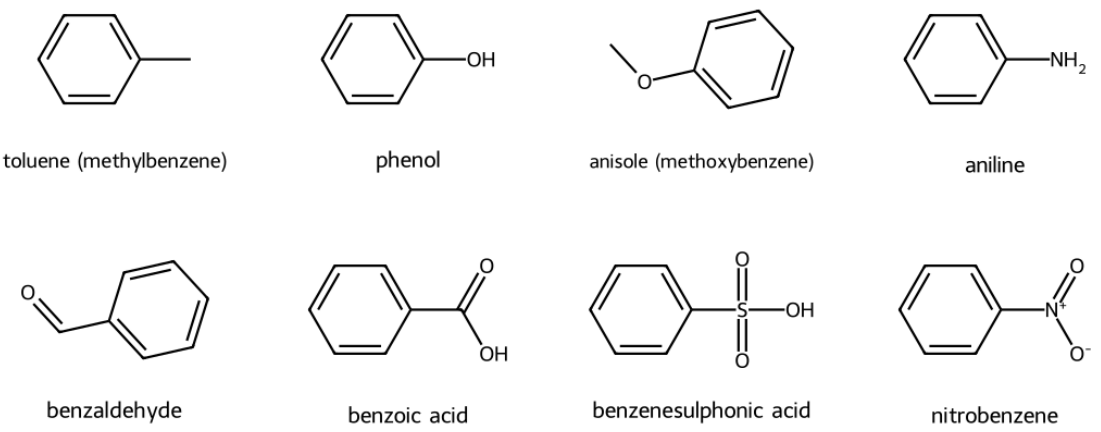
20.2 NOMENCLATURE OF BENZENE DERIVED COMPOUNDS

Q.13 [Long] **How are monosubstituted benzenes named? Give examples.**

Aromatic hydrocarbons that contain only one benzene ring are **monocyclic aromatic hydrocarbons**. When a single substituent is present, it is a **monosubstituted benzene**:

- Most are named with the base name “benzene” and the substituent as a prefix, e.g., chlorobenzene, nitrobenzene, methylbenzene.
- Many have common names that are accepted in IUPAC nomenclature as base names, e.g., phenol, toluene, aniline. The substituent is always at C-1, so its position need not be numbered.

- Some have special names accepted as IUPAC names, e.g., benzenecarboxylic acid, benzenesulphonic acid.

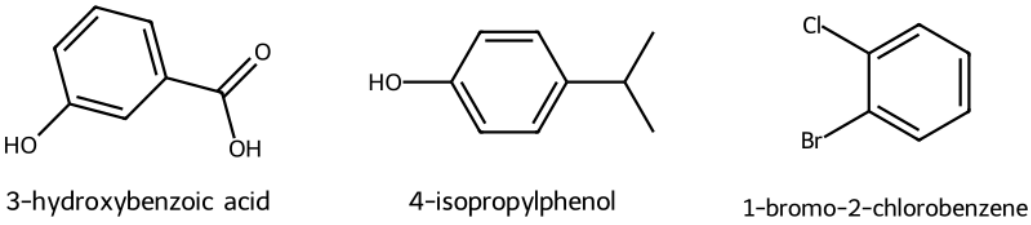


Monosubstituted benzenes

Q.14 [Long] State the rules for naming disubstituted benzenes with examples.

Disubstituted benzenes have two substituents on the ring. Rules:

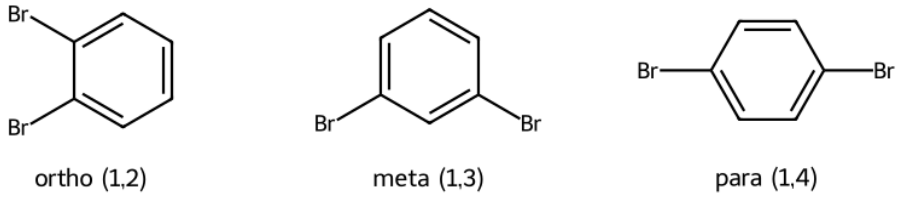
- The carbon carrying the higher priority substituent is C-1; the others are numbered accordingly. Order of priority (left to right): –COOH, –CN, –CHO, –$COCH_3$, –OH, –NH_2, –OR, –R.
- If both substituents have the same priority, the carbon carrying the substituent that comes first in alphabetical order is C-1.



3-Hydroxybenzoic acid: acid has higher priority than alcohol. 4-Isopropylphenol: the phenol carbon is C-1. 1-Bromo-2-chlorobenzene: bromo is before chloro in the alphabet.

Q.15 [Short] What do the terms ortho, meta and para mean?

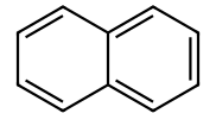
Disubstituted benzenes also have common names based on the relative positions of the two substituents: **ortho (o-)** for 1,2; **meta (m-)** for 1,3; **para (p-)** for 1,4.



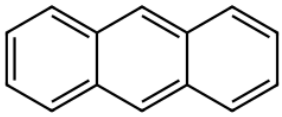
In the textbook's "Did You Know?" figure the third structure (1,4-dibromobenzene) is labelled "ortho (1,3)"; it is the para (1,4) isomer.

Q.16 [Short] What are polycyclic aromatic hydrocarbons? Give examples.

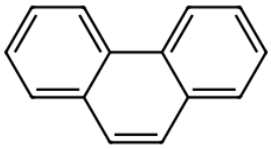
Aromatic hydrocarbons that contain two or more benzene rings in their molecules are **polycyclic aromatic hydrocarbons**.



naphthalene C10H8



anthracene C14H10



phenanthrene C14H10

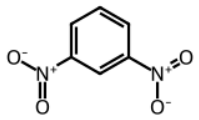
Q.17 [Short] Differentiate between monocyclic and polycyclic aromatic hydrocarbons.

Monocyclic aromatic hydrocarbons contain only one benzene ring, e.g., benzene and toluene. Polycyclic aromatic hydrocarbons contain two or more benzene rings, e.g., naphthalene, anthracene and phenanthrene.

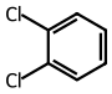
Quick Check 20.2

Q.18 [Short] Give names to following arenes:

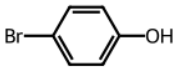
No.	Substituents on the ring	Name
i)	NO_2 at positions 1 and 3	1,3-dinitrobenzene
ii)	Cl at positions 1 and 2	1,2-dichlorobenzene
iii)	OH at 1, Br at 4	4-bromophenol
iv)	CH_3 at 1, Cl at 2 and 4	2,4-dichlorotoluene (2,4-dichloro-1-methylbenzene)
v)	CH_3 at 1, NO_2 at 2, 4 and 6	2,4,6-trinitrotoluene (2-methyl-1,3,5-trinitrobenzene)
vi)	$-COOCH_3$ group	methyl benzoate



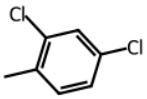
i) 1,3-dinitrobenzene



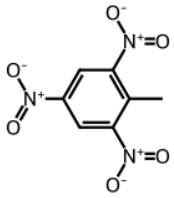
ii) 1,2-dichlorobenzene



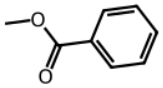
iii) 4-bromophenol



iv) 2,4-dichlorotoluene



v) 2,4,6-trinitrotoluene



vi) methyl benzoate

20.3 REACTIVITY OF BENZENE vs ALKANES AND ALKENES

Q.19 [Long] Compare the reactivity of benzene with alkanes and alkenes.

Alkanes	Alkenes	Benzene
Generally unreactive: non-polar, σ bonds only.	Very reactive: π bond with available π electrons.	Has π electrons, but is very stable due to delocalisation; less reactive than alkenes.
Free radical substitution in UV light.	Electrophilic addition, easily.	Electrophilic substitution, not addition.
$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ (UV)	Cyclohexene + $Br_2 \rightarrow$ 1,2-dibromocyclohexane (no catalyst)	$C_6H_6 + Br_2 \rightarrow C_6H_5Br + HBr$ ($FeBr_3$)

Q.20 [Short] Why does benzene undergo electrophilic substitution rather than electrophilic addition?

In electrophilic substitution a hydrogen atom of the ring is replaced. The reaction passes through an intermediate in which the aromaticity is broken, but it is restored at the end, so the overall stability of the benzene ring (extensive delocalisation of π electrons) is not affected. In electrophilic addition the delocalisation would be permanently broken, so addition is not favourable.

Quick Check 20.3

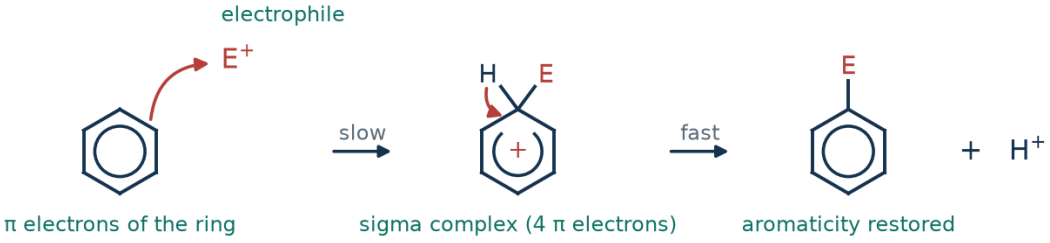
Q.21 [Short] **Although benzene is highly unsaturated yet it does not prefer to undergo addition reactions. Explain.**

The six π electrons of benzene are delocalised over the whole ring, which gives it extra stability (resonance energy 152 kJ mol^{-1}). An addition reaction would break this delocalisation permanently and the stability would be lost, so addition is not favourable. In substitution the aromaticity is broken only in the intermediate and is restored at the end, so benzene prefers electrophilic substitution.

20.3.1 Electrophilic Aromatic Substitution

Q.22 [Long] **Describe the general mechanism of electrophilic aromatic substitution.**

1. **Generation of electrophile:** the electrophile can be a positive ion or the positive end of a polar molecule. It is produced in situ by adding appropriate reagents to the reaction mixture.
2. **Electrophilic attack:** the π electrons of benzene attack the electrophile, forming a σ bond between one ring carbon and the electrophile. This disrupts the aromaticity: only four π electrons are left and a positive charge is spread over five carbon atoms. This intermediate is called the **sigma complex**.
3. **Restoration of aromaticity:** the hydrogen on the sp^3-hybridized carbon is removed by a counter-ion/conjugate base. The C–H bond breaks heterolytically, its electrons go into the π bonding system, and the delocalisation in the ring is restored. This is a rapid step.



General mechanism of electrophilic substitution

Q.23 [Short] **What is a sigma complex?**

It is the intermediate formed when the π electrons of benzene form a σ bond between one ring carbon and the electrophile. The aromaticity is disrupted: only four π electrons remain and a positive charge is spread over the other five carbon atoms. The carbon bonded to the electrophile is sp^3 hybridized.

Q.24 [Long] **Describe the halogenation (bromination) of benzene with its mechanism.**

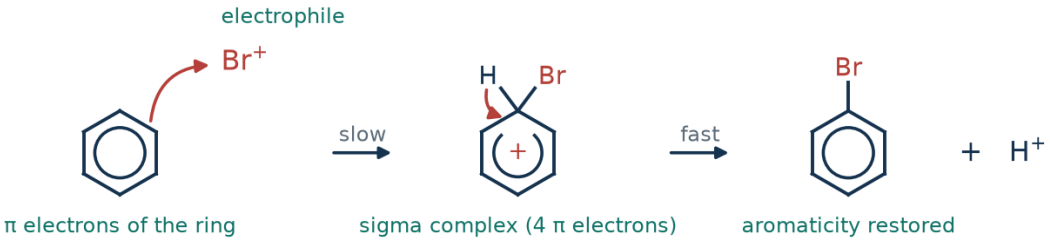
Benzene reacts with bromine and chlorine only in the presence of Lewis acids ($AlCl_3$, $FeCl_3$ or $FeBr_3$), which coordinate to the halogen and generate a strong electrophile.

$$C_6H_6 + Br_2 \rightarrow C_6H_5Br + HBr \text{ (anhydrous } AlBr_3)$$

Step 1, electrophile:

$$Br\text{–}Br + AlBr_3 \rightarrow Br^+ + [AlBr_4]^-$$

Step 2: Br^+ is attacked by the π electrons to form the sigma complex. **Step 3:** $[AlBr_4]^-$ removes the proton, giving bromobenzene and regenerating the catalyst.



$$H^+ + [AlBr_4]^- \rightarrow HBr + AlBr_3$$

The textbook writes that the sigma complex "is then deprotonated by $AlCl_4^-$" in the bromination; with $AlBr_3$ the ion is $[AlBr_4]^-$.

Q.25 [Short] Why does benzene react with chlorine or bromine only in the presence of a Lewis acid?

Benzene is very stable, and the halogen molecule alone is not a strong enough electrophile. Lewis acids such as $AlCl_3$, $FeCl_3$ or $FeBr_3$ coordinate to the halogen and generate the strong electrophile (Cl^+ or Br^+).

Quick Check 20.4

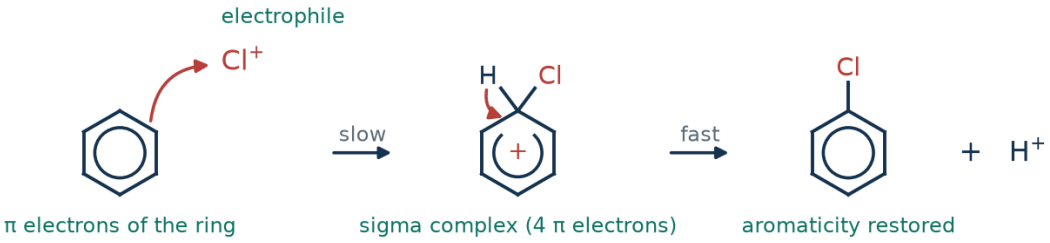
Q.26 [Short] a) Write mechanism of reaction of benzene with chlorine using $AlCl_3$ as catalyst.

$$C_6H_6 + Cl_2 \rightarrow C_6H_5Cl + HCl \text{ (anhydrous } AlCl_3)$$

Step 1, generation of the electrophile: the Lewis acid accepts a lone pair from chlorine and polarises the Cl–Cl bond.

$$Cl^{\delta+}\text{–}Cl^{\delta-} + AlCl_3 \rightarrow Cl^+ + [AlCl_4]^-$$

Step 2, electrophilic attack: the π electrons of benzene attack Cl^+, forming the sigma complex. **Step 3, restoration of aromaticity:** $[AlCl_4]^-$ removes H^+ from the sigma complex.



$$H^+ + [AlCl_4]^- \rightarrow HCl + AlCl_3$$

Q.27 [Long] Explain the nitration of benzene with its mechanism.

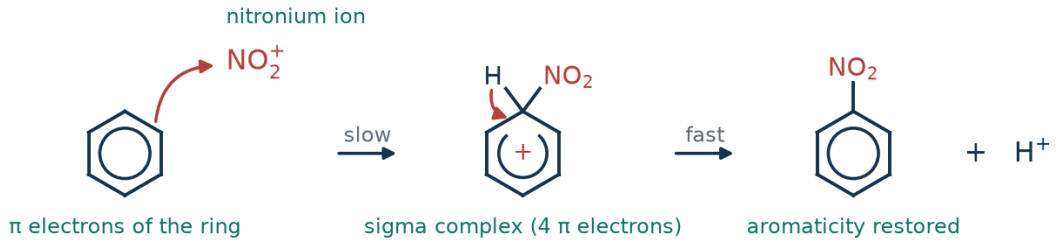
A nitro group is introduced into benzene using a mixture of conc. HNO_3 and conc. H_2SO_4.

$$C_6H_6 + HNO_3 \rightarrow C_6H_5NO_2 + H_2O \text{ (conc. } H_2SO_4\text{, 50 °C)}$$

Formation of electrophile (nitronium ion):

$$H_2SO_4 + HNO_3 \rightleftharpoons NO_2^+ + HSO_4^- + H_2O$$

Attack of electrophile: a pair of electrons from the benzene ring is donated to NO_2^+ to form a covalent bond, causing loss of aromaticity. Then the C–H bond of the substituted carbon breaks and its electrons go back into the π system, restoring aromaticity.



Regeneration of the catalyst:

$$HSO_4^- + H^+ \rightarrow H_2SO_4$$

Concentrated sulphuric acid acts as a catalyst. At higher temperatures more than one nitro group is substituted; some 1,3-dinitrobenzene is formed even at 50 °C.

The learning outcomes print "neutronium ion"; the electrophile is the nitronium ion, NO_2^+.

Q.28 [Short] What happens if the nitration of benzene is carried out at a higher temperature?

At higher temperatures there is a greater chance of more than one nitro group being substituted onto the ring. A certain amount of 1,3-dinitrobenzene is formed even at 50 °C.

Quick Check 20.5

Q.29 [Short] Show how the electrophile is produced in the nitration of benzene from sulphuric acid and nitric acid. Which of these two acts as an acid and which acts as a base?

$$H_2SO_4 + HNO_3 \rightleftharpoons NO_2^+ + HSO_4^- + H_2O$$

Sulphuric acid is the stronger acid: it donates a proton to nitric acid, so **H_2SO_4 acts as an acid**. Nitric acid accepts the proton, so **HNO_3 acts as a base**. The protonated nitric acid then loses water to give the nitronium ion, NO_2^+, the electrophile.

Q.30 [Short] Differentiate between Friedel-Crafts alkylation and acylation.

Friedel-Crafts alkylation	Friedel-Crafts acylation
An alkyl group (R–) is substituted into the ring.	An acyl group (R–C=O) is substituted into the ring.
Reagent: alkyl halide with $AlCl_3$.	Reagent: acyl halide with $AlCl_3$.
Electrophile: carbocation, R^+.	Electrophile: acylium ion, RCO^+.
Product: alkylbenzene, e.g., ethylbenzene.	Product: aryl ketone, e.g., phenylethanone.

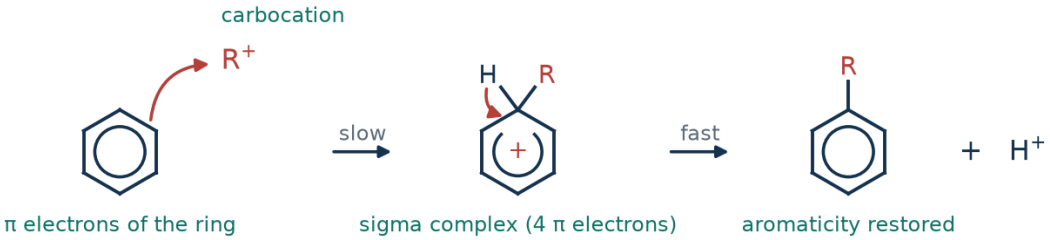
Q.31 [Long] Explain the mechanism of Friedel-Crafts alkylation, including the role of the Lewis acid catalyst.

In Friedel-Crafts alkylation an **alkyl group** is substituted into the benzene ring in the presence of a Lewis acid catalyst.

$$C_6H_6 + CH_3CH_2Cl \rightarrow C_6H_5CH_2CH_3 + HCl\ (AlCl_3, \text{heat})$$

Three-step mechanism:

1. The Lewis acid catalyst ($AlCl_3$) reacts with the alkyl halide, forming a carbocation which acts as the electrophile: $(CH_3)_3C–Cl + AlCl_3 \rightarrow (CH_3)_3C^+ + [AlCl_4]^-$.
2. The π electrons of the aromatic ring attack the carbocation, forming an arenium ion as intermediate. The aromaticity is temporarily lost.
3. Deprotonation of the intermediate re-forms the carbon–carbon double bond and restores aromaticity. The proton forms HCl, regenerating $AlCl_3$.



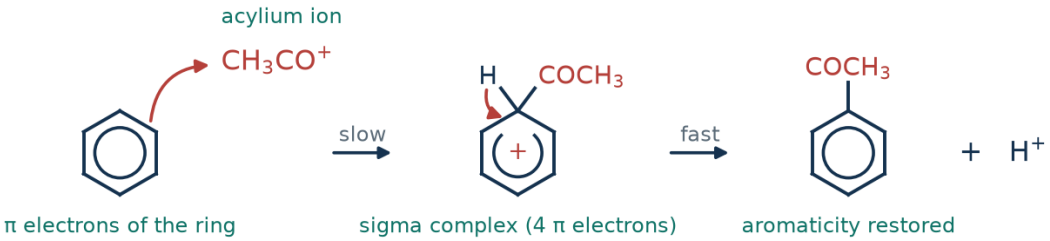
Q.32 [Long] Explain the mechanism of Friedel-Crafts acylation.

In Friedel-Crafts acylation an **acyl group** (an alkyl group attached to a carbonyl group, R–C=O) is substituted into the benzene ring in the presence of a Lewis acid catalyst.

$$C_6H_6 + CH_3CH_2COCl \rightarrow C_6H_5COCH_2CH_3 + HCl\ (AlCl_3, \text{heat})$$

Mechanism:

1. $AlCl_3$ reacts with the acyl halide. A complex is formed and the acyl halide loses a halide ion, forming an **acylium ion**: $RCOCl + AlCl_3 \rightarrow RCO^+ + [AlCl_4]^-$.
2. The π electrons of the aromatic ring attack the acylium ion (RCO^+).
3. The intermediate complex is deprotonated, restoring aromaticity. The proton attaches to a chloride ion forming HCl, and the $AlCl_3$ catalyst is regenerated. An aryl ketone is obtained.



Q.33 [Short] What is the role of the Lewis acid catalyst in Friedel-Crafts reactions?

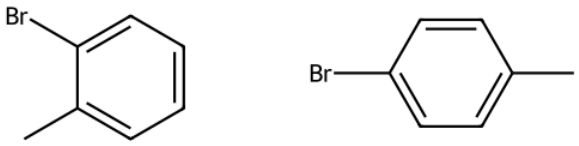
$AlCl_3$ is an electron pair acceptor. It accepts a lone pair from the halogen of the alkyl or acyl halide and removes a halide ion as $[AlCl_4]^-$, producing the electrophile (carbocation or acylium ion). At the end $[AlCl_4]^-$ takes the proton from the intermediate, forming HCl and regenerating $AlCl_3$.

Quick Check 20.6

Q.34 [Short] 1. Write down reactions of methyl benzene with bromine in the presence of $AlBr_3$.

The methyl group is 2,4-directing, so a mixture of 2-bromomethylbenzene and 4-bromomethylbenzene is formed:

$$2C_6H_5CH_3 + 2Br_2 \rightarrow 2\text{-}BrC_6H_4CH_3 + 4\text{-}BrC_6H_4CH_3 + 2HBr\ (AlBr_3)$$



2-bromomethylbenzene 4-bromomethylbenzene

Q.35 [Short] 2. Explain the difference in the reactivity of bromine with benzene and with phenol. Include the type of reaction, the products that form, and any conditions required. Mechanisms for the reactions are not required.

	Benzene	Phenol
Type of reaction	Electrophilic substitution	Electrophilic substitution
Conditions	Liquid bromine with a Lewis acid (halogen carrier) such as $FeBr_3$ or $AlBr_3$	Bromine water at room temperature; no catalyst needed
Products	Bromobenzene and HBr (one Br enters)	2,4,6-tribromophenol (white precipitate) and 3HBr
Reactivity	Less reactive	Much more reactive: –OH activates the 2, 4 and 6 positions

$$C_6H_6 + Br_2 \rightarrow C_6H_5Br + HBr$$

$$C_6H_5OH + 3Br_2 \rightarrow C_6H_2Br_3OH + 3HBr$$

In phenol, a lone pair of electrons on the oxygen atom of the –OH group overlaps with the delocalised π electrons of the benzene ring. This increases the electron density of the ring (the ring is activated), especially at the 2, 4 and 6 positions, so it attracts and polarises electrophiles such as bromine much more strongly. Benzene has a lower π electron density and needs a Lewis acid (halogen carrier) to generate the electrophile.

The textbook states that –OH directs incoming groups to the 2, 4 and 6 positions; the bromine-water conditions and the white precipitate of 2,4,6-tribromophenol are from the phenol chapter (Ch. 22).

Q.36 [Short] 3. Explain the difference in the reactivity of bromine with benzene and with ethyl benzene. Include the type of reaction, the products that form, and any conditions required. Mechanisms for the reactions are not required.

Both react by electrophilic substitution with bromine and a Lewis acid ($AlBr_3$ or $FeBr_3$) in the absence of UV light. Benzene gives bromobenzene. Ethylbenzene reacts faster, because the ethyl group donates electrons to the ring and activates it; it is 2,4-directing, so a mixture of 1-bromo-2-ethylbenzene and 1-bromo-4-ethylbenzene is formed.

$$C_6H_6 + Br_2 \rightarrow C_6H_5Br + HBr\ (AlBr_3)$$

$$2C_6H_5C_2H_5 + 2Br_2 \rightarrow 2\text{-}BrC_6H_4C_2H_5 + 4\text{-}BrC_6H_4C_2H_5 + 2HBr\ (AlBr_3)$$

Q.37 [Short] 4. Which of 1,1-dimethyethyllbenzene and methyl benzene is easier to form by Friedel-Craft reaction? Explain based on the stability of carbocation.

1,1-Dimethylethylbenzene is easier to form. The electrophile in Friedel-Crafts alkylation is a carbocation. For 1,1-dimethylethylbenzene it is the tertiary carbocation $(CH_3)_3C^+$, which is stabilised by three electron-donating methyl groups and forms easily from $(CH_3)_3CCl$ and $AlCl_3$. For methylbenzene it would be the methyl carbocation, CH_3^+, which has no alkyl groups and is the least stable carbocation, so it is difficult to form.

20.4 ORIENTATION IN ELECTROPHILIC SUBSTITUTION REACTIONS

Q.38 [Short] What is meant by orientation in electrophilic substitution?

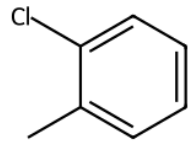
Any group already attached to the benzene ring affects the electrophilic substitution reaction and determines the position (orientation) at which the incoming electrophile enters. Substituents direct the incoming electrophile in two ways: ortho-para directing and meta directing.

Q.39 [Long] What are ortho-para directing (activating) groups? Explain with an example.

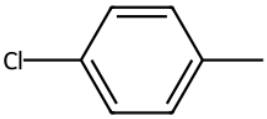
A group that makes the ring **more reactive** than benzene and directs the incoming electrophile to the ortho (1,2) and para (1,4) positions is an **ortho-para director (activating group)**. Ortho and para products dominate, with a slight preference for para; the meta product is an extremely minor by-product. Examples: –SR, –R, –OH, –OR, $–NR_2$, –SH. These groups can donate electrons to the benzene ring.

Example: the methyl group of methylbenzene is 2,4-directing. With chlorine and $AlCl_3$ (or iron), in the absence of UV light, at room temperature, a mixture of 2-chloromethylbenzene and 4-chloromethylbenzene is formed. The –OH group is also a ring activator; incoming groups go into the 2, 4 and 6 positions.

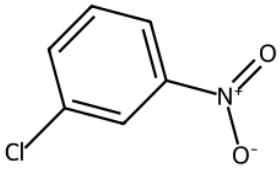
$$2C_6H_5CH_3 + 2Cl_2 \rightarrow 2\text{-}ClC_6H_4CH_3 + 4\text{-}ClC_6H_4CH_3 + 2HCl \text{ (anhydrous } AlCl_3)$$



2–chloromethylbenzene



4–chloromethylbenzene



3–chloronitrobenzene (meta)

Q.40 [Long] What are meta directing (deactivating) groups? Explain with an example.

A group that makes the ring **less reactive** than benzene and directs the incoming electrophile to the meta (1,3) position only is a **meta director (deactivating group)**. Such groups are mostly electron-withdrawing groups that destabilize the arenium intermediate. Examples: –CN, C=O, $–NO_2$, $–CO_2–$. **Example:** chlorination of nitrobenzene (Cl_2, $AlCl_3$) occurs at the meta position, giving meta-chloronitrobenzene.

Q.41 [Short] Differentiate between ortho-para directing and meta directing groups.

Ortho-para directing groups	Meta directing groups
Activating: ring more reactive than benzene.	Deactivating: ring less reactive than benzene.
Donate electrons to the ring.	Withdraw electrons from the ring; destabilize the arenium intermediate.
Product: mainly 1,2 and 1,4 isomers.	Product: mainly the 1,3 isomer.
–R, –OH, –OR, $–NR_2$, –SR, –SH	$–NO_2$, –CN, C=O, $–CO_2–$

Q.42 [Short] **Why must the chlorination of the ring of methylbenzene be done in the absence of UV light?**

Ring substitution is an electrophilic substitution that needs a Lewis acid ($AlCl_3$ or iron) at room temperature. In UV light chlorine forms free radicals, which substitute the hydrogen atoms of the methyl side chain instead (free radical substitution, e.g., $C_6H_5CH_3 \rightarrow C_6H_5CCl_3$).

The textbook only states "absence of UV light"; the side-chain reaction is from Chapter 19 (exercise Q3 a).

Quick Check 20.7

Q.43 [Short] **a) Explain why phenol can be nitrated using milder conditions than for benzene.**

The –OH group is an activating group: a lone pair on its oxygen is donated into the ring, increasing the electron density of the ring. The activated ring is attacked by the electrophile much more easily, so dilute nitric acid at room temperature is enough. Benzene needs conc. HNO_3 with conc. H_2SO_4 at about 50 °C.

Q.44 [Short] **b) How would you prepare 3-chloronitrobenzene from benzene?**

Nitrate first, then chlorinate. The nitro group is meta directing, so chlorine enters the 3-position.

$$C_6H_6 + HNO_3 \rightarrow C_6H_5NO_2 + H_2O \text{ (conc. } H_2SO_4\text{, 50 °C)}$$
$$C_6H_5NO_2 + Cl_2 \rightarrow 3\text{-}ClC_6H_4NO_2 + HCl \text{ } (AlCl_3)$$

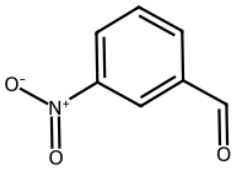
If chlorination were done first, the product would be the 2- and 4-isomers.

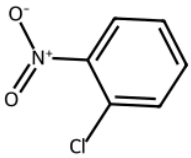
Q.45 [Short] **c) Which products are expected to form in majority when the following are nitrated with conc. sulphuric acid and nitic acid? Give reactions. i. benzaldehyde ii. chlorobenzene iii. phenol iv. Benzene carboxylic acid**

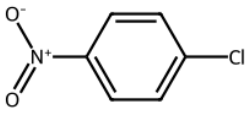
	Compound	Group	Major product(s)
i.	benzaldehyde	–CHO (C=O), meta directing	3-nitrobenzaldehyde
ii.	chlorobenzene	–Cl, ortho-para directing	2-nitrochlorobenzene and 4-nitrochlorobenzene
iii.	phenol	–OH, ortho-para directing	2-nitrophenol and 4-nitrophenol
iv.	benzenecarboxylic acid	–COOH (C=O), meta directing	3-nitrobenzoic acid

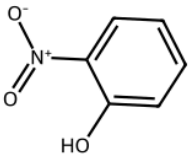
$$C_6H_5CHO + HNO_3 \rightarrow 3\text{-}O_2NC_6H_4CHO + H_2O$$
$$2C_6H_5Cl + 2HNO_3 \rightarrow 2\text{-}O_2NC_6H_4Cl + 4\text{-}O_2NC_6H_4Cl + 2H_2O$$
$$2C_6H_5OH + 2HNO_3 \rightarrow 2\text{-}O_2NC_6H_4OH + 4\text{-}O_2NC_6H_4OH + 2H_2O$$

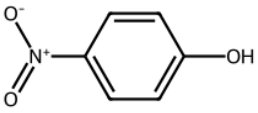
$$C_6H_5COOH + HNO_3 \rightarrow 3\text{-}O_2NC_6H_4COOH + H_2O$$

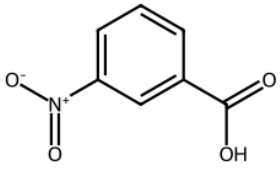

3-nitrobenzaldehyde


2-nitrochlorobenzene


4-nitrochlorobenzene


2-nitrophenol


4-nitrophenol


3-nitrobenzoic acid

Halogens are not in the textbook's list of directing groups; chlorine is ortho-para directing (general knowledge).

20.4.1 Combustion

Q.46 [Short] Describe the combustion of benzene. What is the smoky flame test?

When benzene is burnt in excess of air it is completely oxidized to CO_2 and H_2O. It burns with a smoky flame, which is used as a test for aromatic compounds.

$$2C_6H_6 + 15O_2 \rightarrow 12CO_2 + 6H_2O$$

20.4.2 Side Chain Oxidation in Benzene

Q.47 [Long] Explain side-chain oxidation of alkylbenzenes, including the formation of benzoic acid.

Oxidation that takes place at an alkyl group directly attached to a benzene ring is called **side-chain oxidation**. Alkylbenzenes are readily oxidized by alkaline $KMnO_4$, acidic $K_2Cr_2O_7$ or dilute HNO_3. The alkyl group is oxidized to a carboxylic acid group, whatever its size, and the benzene ring stays intact.

$$C_6H_5CH_3 + 3[O] \rightarrow C_6H_5COOH + H_2O \text{ (heat)}$$

Condition: there must be at least one **benzylic hydrogen** (a hydrogen on the carbon attached to the ring). Tertiary carbons attached to the ring have no benzylic hydrogen and do not react. Both alkanes and benzene are stable to oxidation, but the benzylic carbon directly attached to the ring is more reactive and more susceptible to oxidation.

study-sapiens.com

Q.48 [Short] What is a benzylic hydrogen? Why is it necessary for side-chain oxidation?

A benzylic hydrogen is a hydrogen atom attached to the carbon that is directly bonded to the benzene ring. Side-chain oxidation takes place only when at least one benzylic hydrogen is present; an alkyl group attached through a tertiary carbon (no benzylic hydrogen) gives no reaction.

Q.49 [Short] How is side-chain oxidation used as an identification test?

Oxidation by $KMnO_4$ is used as an identification test: the purple colour of $KMnO_4$ is discharged when it is consumed. So benzene (no reaction), benzenes with primary or secondary alkyl groups (colour discharged) and with tertiary alkyl groups (no reaction) can be differentiated.

20.4.3 Addition Reactions of Benzene

Q.50 [Long] Explain the hydrogenation of benzene, including the role of the metal catalyst.

Hydrogenation of benzene and other arenes involves the addition of 1, 2 or 3 moles of hydrogen to the ring carbons. Complete hydrogenation of benzene gives cyclohexane, a useful solvent and a precursor for other chemicals.

$$C_6H_6 + 3H_2 \rightarrow C_6H_{12} \text{ (cyclohexane) (Ni, about 150 °C, high pressure)}$$

$$C_6H_5CH_3 + 3H_2 \rightarrow C_6H_{11}CH_3 \text{ (methylcyclohexane)}$$

The reaction destroys the electron delocalisation of the ring, because those electrons are used to form bonds with the new hydrogen atoms. Although it is exothermic overall (strong new C–H bonds), there is a high activation barrier. So the same finely divided nickel catalyst as for alkenes is used at a similar temperature (around 150 °C), but with a higher pressure.

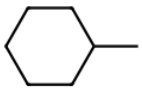
Q.51 [Short] Describe the addition of halogens to benzene.

Chlorine and bromine add in the same way. In the absence of a catalyst, UV light and heat are sufficient for complete halogenation.

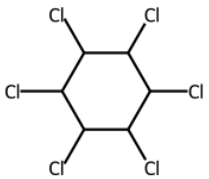
$$C_6H_6 + 3Cl_2 \rightarrow C_6H_6Cl_6 \text{ (1,2,3,4,5,6-hexachlorocyclohexane) (UV, heat)}$$



cyclohexane



methylcyclohexane



1,2,3,4,5,6-hexachlorocyclohexane

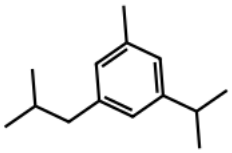
Q.52 [Short] **Differentiate between the reaction of benzene with chlorine in the presence of $AlCl_3$ and in the presence of UV light.**

With $AlCl_3$ (anhydrous)	With UV light and heat
Electrophilic substitution.	Addition.
One H is replaced by Cl.	Three Cl_2 molecules add to the ring.
Product: chlorobenzene + HCl; the ring stays aromatic.	Product: 1,2,3,4,5,6-hexachlorocyclohexane; the delocalisation is destroyed.

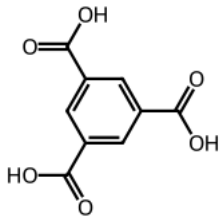
Quick Check 20.8

Q.53 [Short] **a) Predict the product formed when: i. the compound given here is oxidized using acidified $KMnO_4$ or $K_2Cr_2O_7$. ii. Bromine is added to chlorobenzene in the presence of $AlBr_3$. iii. Nitrobenzene is reduced in the presence of Ni at high temperature and pressure.**

i. The ring carries three alkyl groups at positions 1, 3 and 5, and each has at least one benzylic hydrogen. Each alkyl group is oxidized to –COOH whatever its size, so the product is **benzene-1,3,5-tricarboxylic acid**.



compound given



benzene-1,3,5-tricarboxylic acid

ii. Electrophilic substitution; chlorine is ortho-para directing, so a mixture of **1-bromo-2-chlorobenzene and 1-bromo-4-chlorobenzene** (with HBr) is formed.

iii. With hydrogen and nickel at high temperature and pressure the benzene ring is hydrogenated to a cyclohexane ring; the nitro group is also reduced to $–NH_2$. The product is **cyclohexanamine**, $C_6H_{11}NH_2$ (via phenylamine).

$$C_6H_5NO_2 + 6H_2 \rightarrow C_6H_{11}NH_2 + 2H_2O \text{ (Ni, heat, pressure)}$$

i. The structure drawn has methyl, 1-methylethyl and 2-methylpropyl groups, but its printed label is "3-Ethyl-5-propyltoluene"; either way the product is the same tricarboxylic acid. iii. The textbook describes only hydrogenation of the ring; reduction of the nitro group is general knowledge. If only the ring is considered, the product would be nitrocyclohexane.